Design of 3 Bit Gray To Binary Converter

Group Number:- 13

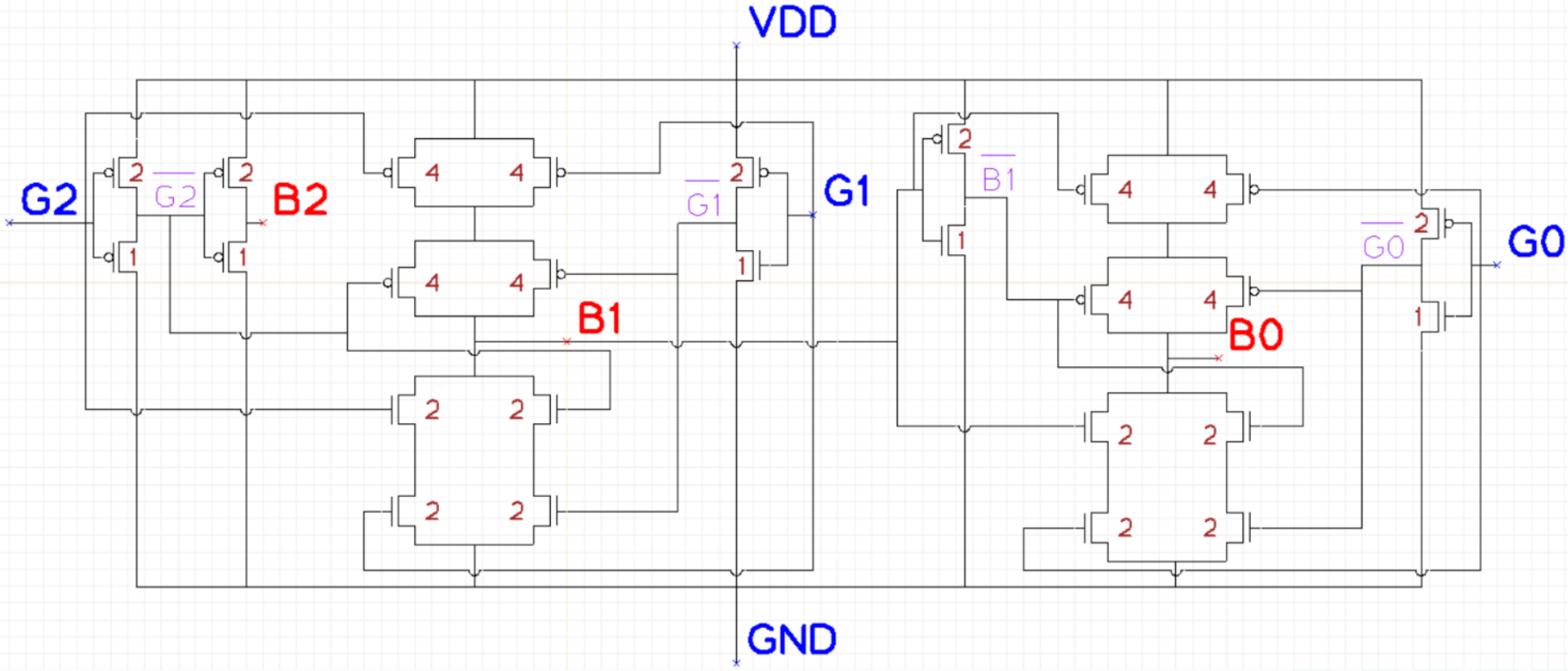
Guided By :-
Dr. Anuj Grover
Sameer Tiwari
Vishal Vishwakarma

Group Members:-

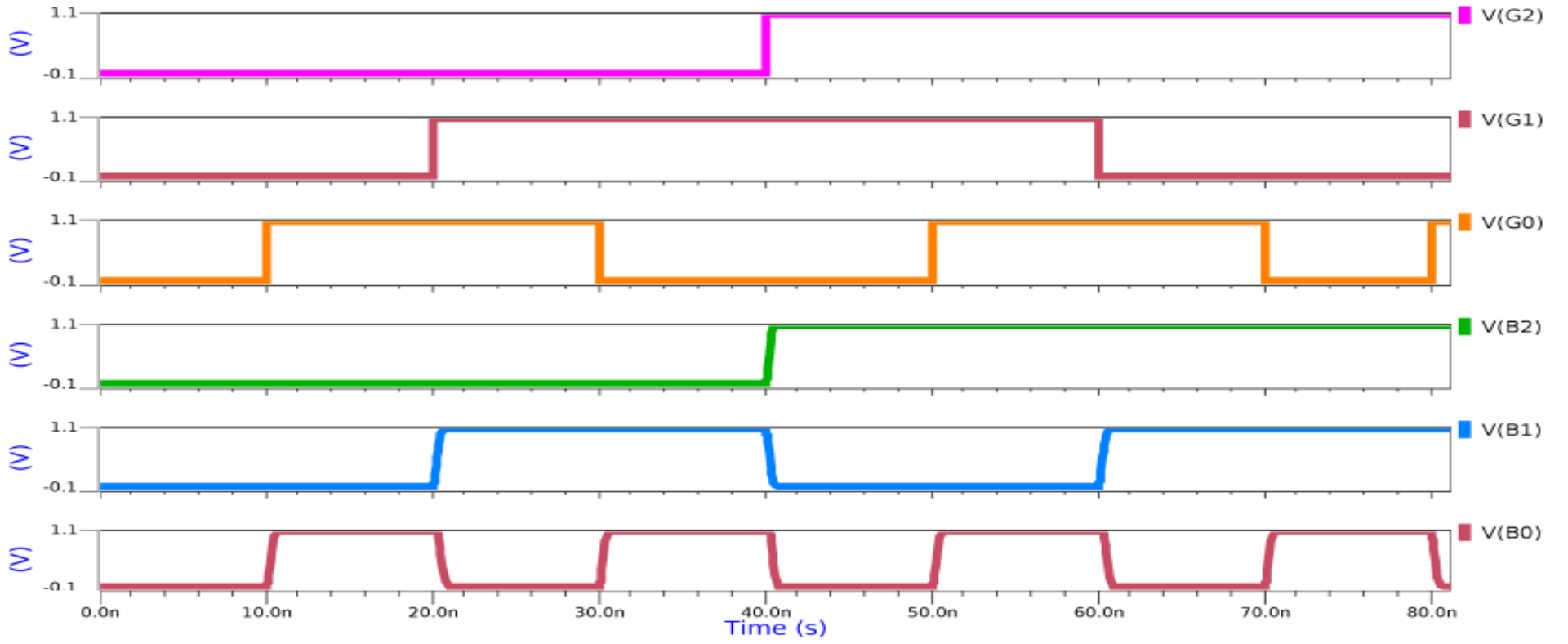
- Nishant Gupta (2024393)
- Pranav Ramanathan (2024424)
- Sushruth S G (MT26215)
- Hrithik R (MT26165)

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY **DELHI**

Schematic + Sizing



Stimuli For Verification & Verification Plan



Stimuli For Verification & Verification Plan

G2	G1	G0	B2	B1	B0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	1	0	1	0
0	1	0	0	1	1
1	1	0	1	0	0
1	1	1	1	0	1
1	0	1	1	1	0
1	0	0	1	1	1

Corner	Parameter	B2	B1	B0
SS	**RISE**	3.6687E-10	4.9978E-10	4.7821E-10
SS	**FALL**	3.8011E-10	4.8771E-10	5.2637E-10
SS	**TPLH**	2.0061E-10	2.4898E-10	2.4400E-10
SS	**TPHL**	2.0295E-10	2.6270E-10	2.3184E-10
SS	**DELAY**	2.0178E-10	2.5629E-10	2.3792E-10
FF	**RISE**	2.6600E-10	3.5481E-10	3.2276E-10
FF	**FALL**	2.3605E-10	3.0197E-10	3.5089E-10
FF	**TPLH**	1.3637E-10	1.4521E-10	1.6474E-10
FF	**TPHL**	1.3175E-10	1.7735E-10	1.3548E-10
FF	**DELAY**	1.3406E-10	1.6128E-10	1.5011E-10

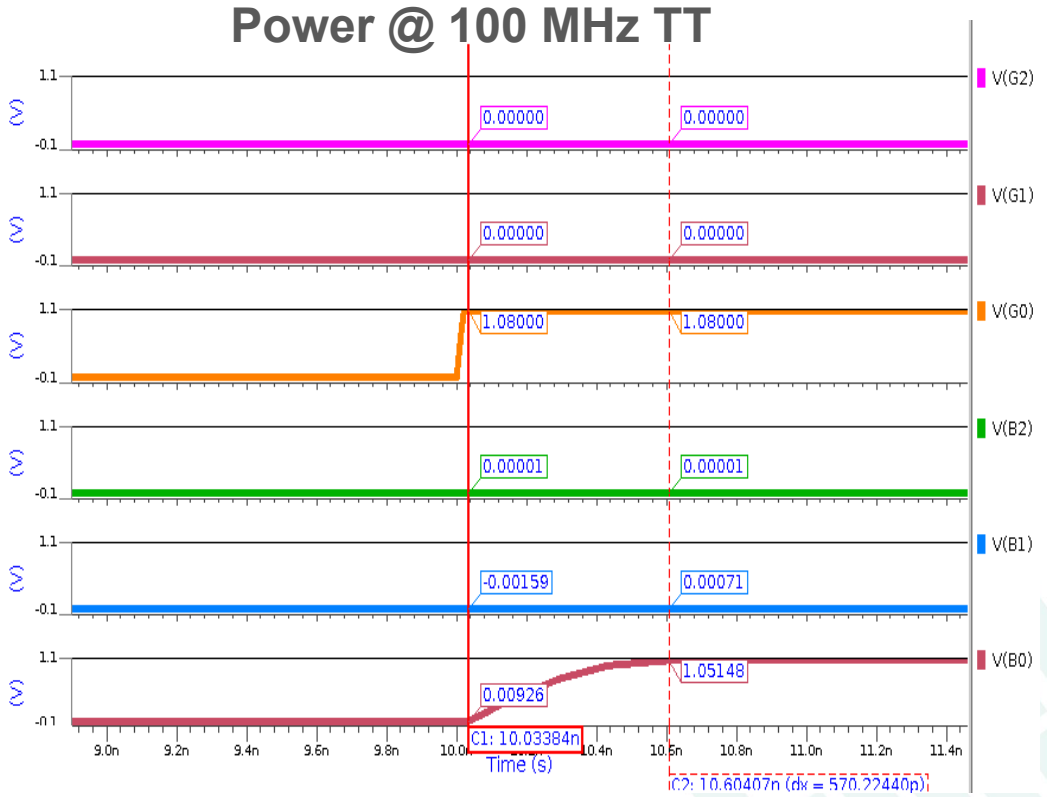
Stimuli For Verification & Verification Plan

TT

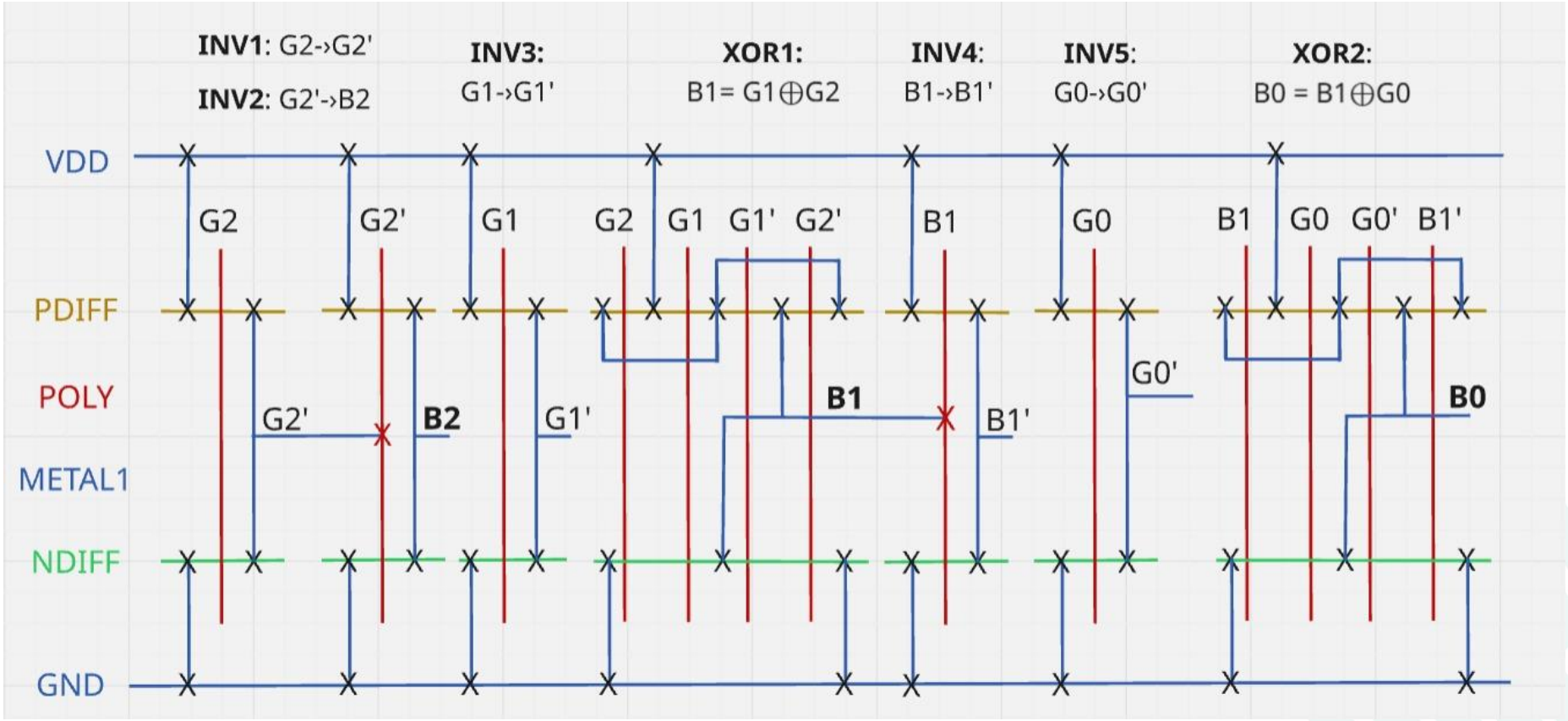
Parameter	B0	B1	B2
RISE	3.8506E-10	4.1958E-10	3.1630E-10
FALL	4.2690E-10	3.8666E-10	3.1692E-10
TPLH	2.0325E-10	2.1547E-10	1.6533E-10
TPHL	1.7684E-10	1.9089E-10	1.6504E-10
DELAY	1.9004E-10	2.0318E-10	1.6519E-10

Power

Parameter	B0
P_STATIC_AVG	7.0115E-07
P_DYNAMIC	3.2697E-05



Stick Diagram



Future Work + Work Distribution

Completed Work:

- Schematic & Sizing :- Nishant Gupta
- Verification & Pre layout :- Sushruth S G & Hrithik R
- Stick Diagram :- Pranav Ramanathan

Future Work:

- Layout and PEX
- Pre Layout vs Post Layout Analysis
- Power Analysis